

## GIS Practical 2 Worksheet

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### GenAI Statement:

The purpose of this assessment is to develop your cartographic and GIS skills – these involve practical skills, understanding of core concepts, and your existing knowledge.

You should not use (or need to use) GenAI for questions 1 and 2. You must demonstrate your core skills and knowledge, making specific reference to the spatial datasets that you are working with in QGIS.

You should use GenAI for question 3, where you will **critically evaluate** the outputs of GenAI when prompted to discuss the suitability of projection systems in your case study.

You should only use GenAI as directed above during the assessment; you should demonstrate your core skills and knowledge.

### Academic integrity:

By submitting this assignment, you are certifying that the work you are submitting is your own and has been done by you and not in consultation with anyone else. You, or anyone else, have not submitted this work for assessment already, either at University College Cork or elsewhere. You have read and understood University College Cork's **Academic Integrity for Examinations and Assessments Policy** (available [here](#)). You understand that the work you are submitting may be checked for academic misconduct and plagiarism, in line with the policy, and that this can result in a reduction in marks or other penalties.

### Guidelines:

You should complete the questions in the worksheet (30% of your grade) and either insert your mapwork (worth 70% of your grade) on the last page or upload your map as a separate file through Canvas.

## GIS Practical 2 – Projections

1. (Approximately 100 words) What projection systems have you worked with when completing this practical?

Zooming out to the full extent of the globe, change the project properties to **identify** – use the measure line tool in QGIS to support your observations (for example, measure the length of Ireland north-south in different projection systems; try using the measure area tool to check for differences in area calculations).

In this practical, I worked with the UTM 36S projection that was used to map the area of the fires in South Africa originally. I also worked with the World Mercator and Robinson projections when comparing the area of the largest fire to the UTM 36S projection. Shown in the final map are the areas of the largest fire in the UTM 36S projection and the Robinson projection.

2. Drawing from the information you've included above, **explain** why this distortion is happening – what are the properties of the projection systems (area, distance, shape, direction) and why do these properties explain the distortion that you're seeing? Use [EPSG.org](https://epsg.org) to support your points (approximately 150 words)

The UTM 36S projection is a transverse Mercator projection made to most accurately represent the areas “between 30°E and 36°E, southern hemisphere between 80°S and equator, onshore and offshore. Burundi. Eswatini (Swaziland). Kenya. Malawi. Mozambique. Rwanda. South Africa. Tanzania. Uganda. Zambia. Zimbabwe.” This projection is most accurate for the fires in South Africa because it doesn't have to worry about distortion for regions outside of those detailed. ([source](#))

The Robinson projection is a whole world projection that compromises on area and angles between other projection systems, without keeping any one attribute entirely accurate. The projection tends to undershoot areas around the equator and then overshoot them near the poles. Because our fires are somewhat close to the equator, we see the Robinson projection as having the smaller area. ([source](#))

3. You should be confident that you can outperform Generative AI in doing spatial analysis. This is because you have **intelligence**; you have practical skills, you have core knowledge, and you know how to apply them. You should also be able to critically assess the accuracy of the information that Generative AI presents. Here, let's work on your **critical thinking and digital literacy skills**. We'll do this by assessing GenAI's ability to discuss different projection systems and their uses.

On the page 4 there is an example of the type of answer that Chat GPT returns when asked to compare the two projection systems that you have been working with (EPSG 32736/UTM 36S and 54030/World Robinson). Figure 1 is a prompt for general overview, figure 2 is a prompt for more specific advice on working with a South African study area.

Using either the example in these figures **or** your own prompt (delete figure 1 and 2, paste in your own prompts) and your mapwork project, critically assess **how accurate** is the information that GenAI is presenting? Identify **at least two** appropriate sources to cross-check the information that you've been given by GenAI.

You should then use the [CRAAP Test](#) to **evaluate** the information that you have collected (from GenAI and your two sources) and **critically assess** what types of information are required to make informed and responsible decisions about projection systems when doing spatial analysis (this should be approximately 350 words)/

After cross-checking the information given by Gen AI, I wasn't able to find any information given that was plainly incorrect. I did notice that it tends to give very vague advice, or simply repeat the same thing twice but with different wording.

I found all the information listed about the two projection systems to be accurate according to [1], [2], and [3].

I also checked the info given for advice on South African tasks checking with [4] & [5] as well.

I wasn't very surprised by this, as most Gen AI now have the capability of citing their sources and generally get quite important details correct.

Sources:

1. <https://pro.arcgis.com/en/pro-app/latest/help/mapping/properties/transverse-mercator.htm>
2. <https://pro.arcgis.com/en/pro-app/latest/help/mapping/properties/robinson.htm>
3. <https://epsg.io/32736>
4. <https://epsg.io/32735>
5. <https://www.fig.net/organisation/comm/5/library/reports/wonnacott.pdf>

Judgment by the CRAAP Test:

Currency:

The information given seemed to be up to date and accurate. It even got correct that the current datum has changed in South Africa and gave the correct new one. No sources were given though I could have asked for them.

For the sources I found, I wasn't able to verify when they had been written, but they seem to be currently updated as they're used by current geographers. Also, the info about map projections is quite old and usually a projection doesn't change once it's been published.

Relevance:

I found the Gen AI information to be relevant and helpful. Though, I don't love the way that ChatGPT writes as I think it's overly confirmatory and dances around the point a lot. I also think a downside of the models is that they're trained to confirm biases and have trouble telling you you're wrong. Also, it tends to repeat things often and that can hurt its ability to give concise information that doesn't require a lot of output.

The relevancy of the info on the sources is going to depend on the sources, as for more niche projections like the UTM 36S, it can be quite hard to find a lot of info online.

Authority:

While there is no authorship with a LLM, it does proclaim things very certainly, which can be a downside when it's just a prediction model and has errors occasionally. The writing tends to be sound and without grammatical errors though.

For the authority on the other sources, I'd expect a level of certainty with information coming from the official EPSG website, as well as academic sources. I'm more likely to trust that than info given by GenAI without a source.

Accuracy:

In this task, I couldn't find anything given by ChatGPT that was inaccurate. I think it tends to do better at simple informative tasks than reasoning-based tasks.

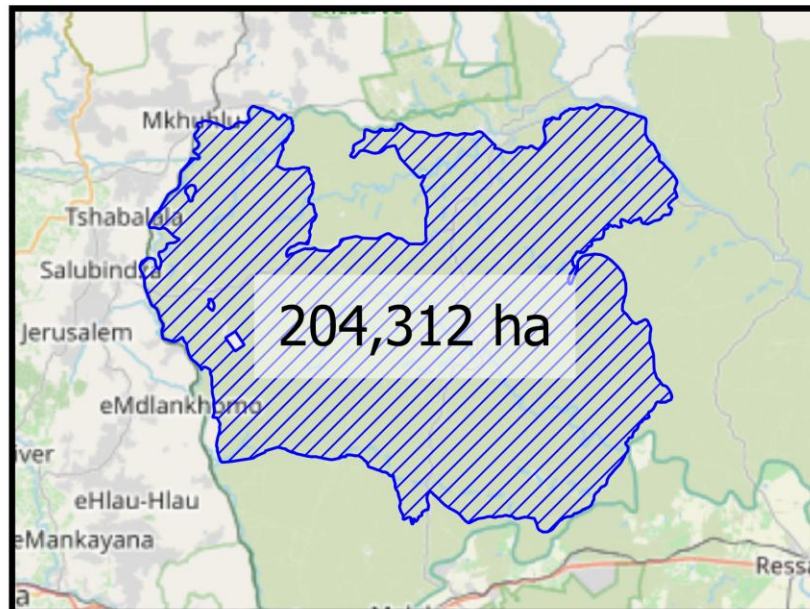
**Purpose:**

This one is quite simple, as most information about projections seems to just be to inform geographers and is often maintained by academics or government agencies. For the EPSG database, it is run by several organizations and therefore has a lot of oversight to maintain correct information.

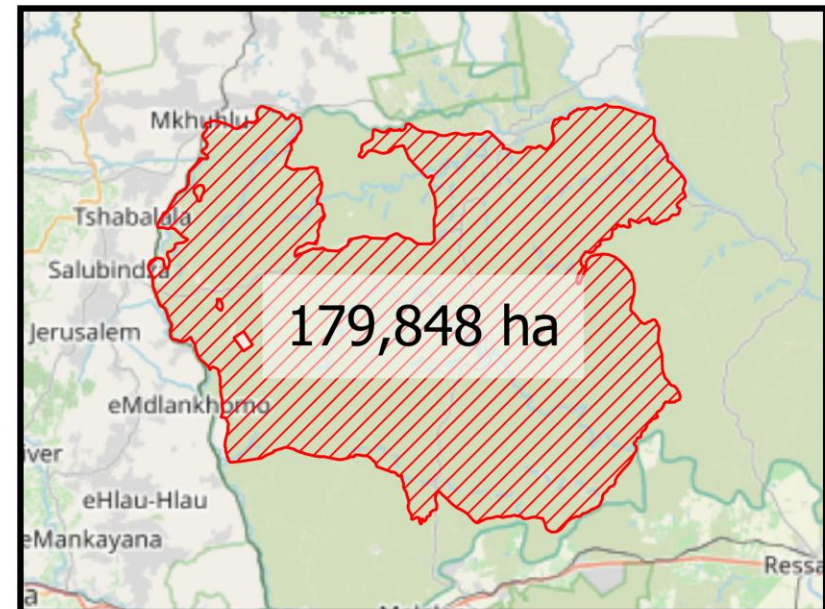
Overall, I think that Gen AI can be very helpful when looking for a quick overview of different projection systems as that information is quite readily available for Gen AI to scrape accurately. When choosing a projection to work with on a project, I would still rather confirm with primary/governmental sources as that would be quite embarrassing to choose incorrectly.

# Area of the Largest Fire in Kruger National Park, South Africa (1992-2001) Across Projection Methods

## UTM 36S Projection



## Robinson Projection



0 7.5 15 km

Mapwork by Max Warriner (2026)

